What is Cypher?

It's SQL for graph databases.

Graph databases store connections: who paid whom, who follows whom, what ships where.

Frontier models get Cypher right less than half the time. Claude Opus 5:

50%

- Fraud rings (FinCEN money transfers)
- Social networks (Twitter)
- Recommendations (movies)
- Supply chains (Northwind orders)

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Your data is a graph. Your questions wait in a ticket queue.

“Which countries received the most money from Singapore?”

```
MATCH (f:Filing) WHERE f.originator_iso = "SGP"
MATCH (f)-[:BENEFITS]->(e:Entity)-[:COUNTRY]->(c:Country)
WITH c.name AS country, count(f) AS filings
ORDER BY filings DESC RETURN country, filings LIMIT 5
```

01

Analyst asks.

02

Engineer writes this.

03

Answer in 2 days.

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Why Cypher

We tested everyday business tasks where a computer can check the answer.

WHAT GOES IN → WHAT COMES OUT	KIND	SMALL, OPEN	FRONTIER
QUESTION → GRAPH DATABASE QUERY	Write code	48	47
CONTRACT CLAUSE → CLAUSE TYPE	Pick a label	69	74
PRODUCT WEB PAGE → PRODUCT DATA FORM	Fill a form	77	70
SUPPORT TICKET → REASON FOR CONTACT	Pick a label	87	88
REQUEST → APP COMMAND WITH SETTINGS	Fill a form	92	92
REQUEST → RIGHT TOOL OUT OF 100	Pick a label	98	95
BROKEN CODE → FIXED CODE	Write code	99	96

Everywhere else, models already score 70 to 99. Graph queries are where even frontier models fail half the time.

Footnote: Small, open = GLM 5.3 Flash. Frontier = Claude Opus 5 (graph query), Kimi K3 (clause type, reason for contact, right tool), GLM 5.3 (product data, app command, fixed code).

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Tech stack

Hugging Face: the public neo4j/text2cypher-2025v1 test set

Langfuse: traces every step: query, database error, repair, cost

Neo4j live graphs: the grader. Every query runs on a real database and the rows are compared.

Fireworks Training API: LoRA fine-tuning (SFT) on Muse Glimmer 30B, the biggest jump: +10 points

Fireworks serverless: every open model through one API

- GLM 5.3 Flash
- Kimi K3
- DeepSeek V4.1
- Muse Glimmer 30B
- NVIDIA Nemotron

Without Fireworks' Training API, fine-tuning a 30B model in one day would not have been possible.

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Benchmark

50.5 → 68.5

Frontier-class model → our system. Same 200 held-out questions, every query run on a live graph.

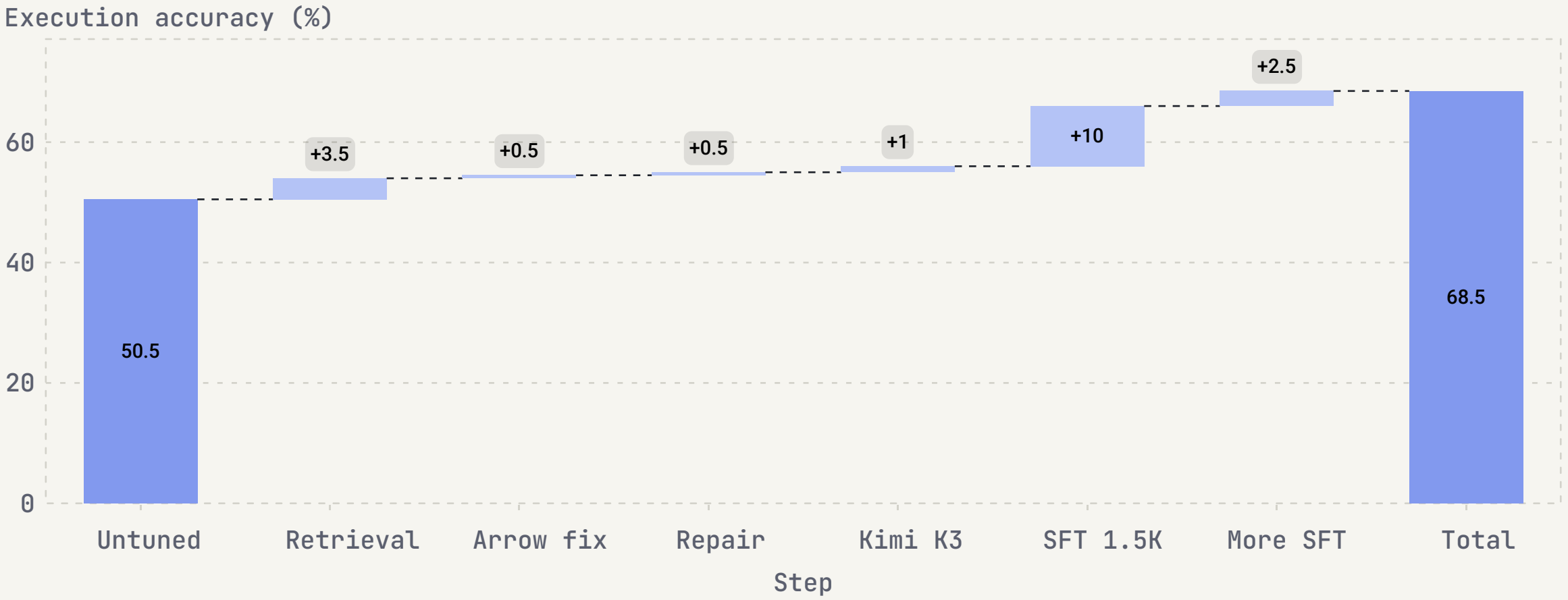
	FRONTIER	OURS
CORRECT OUT OF 100	50.5	68.5
COST PER 1,000 QUESTIONS	$9.00	$0.77
RESPONSE TIME	2.1 s	1.2 s

Footnote: Frontier on the 200 = Kimi K3, run by us. Claude Sonnet 4.6 published 56.6 on another subset. Cost and latency: Claude Opus 5 vs our open-model harness on the same 100 questions.

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What moved the score

50.5 → 68.5. Fine-tuning (SFT) gave the biggest jump: +10.



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What we learned

Price is not accuracy. A $0.24 open model beat $9 Claude.

SFT was the jump. +10 points.

Teach first, then reward. RFT can only reward answers the model already finds. On hard queries it found none, so SFT beat it by 8 points.

The harness buys trust. Broken queries fell from 2.2% to 0.1%.

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Value

$25M improvement

% value created / cost improvement: input needed

$2.6B graph database market, growing ~25% a year

Next: RFT on top of the fine-tuned model. Low-confidence answers go to human review.

Footnote: Assumes 10 graph questions per analyst per week, 48 weeks, 20 minutes of engineer time each, $75/hr. Market: Expert Market Research 2025, $2.58B.

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Reproduce

- Code: https://github.com/raheeldeal/fireworks-hackathon
- Results: https://github.com/raheeldeal/fireworks-hackathon/blob/main/hackathon/cypher/RESULTS_FINAL.md
- Test set: https://huggingface.co/datasets/neo4j/text2cypher-2025v1
- Live graphs: https://demo.neo4jlabs.com

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